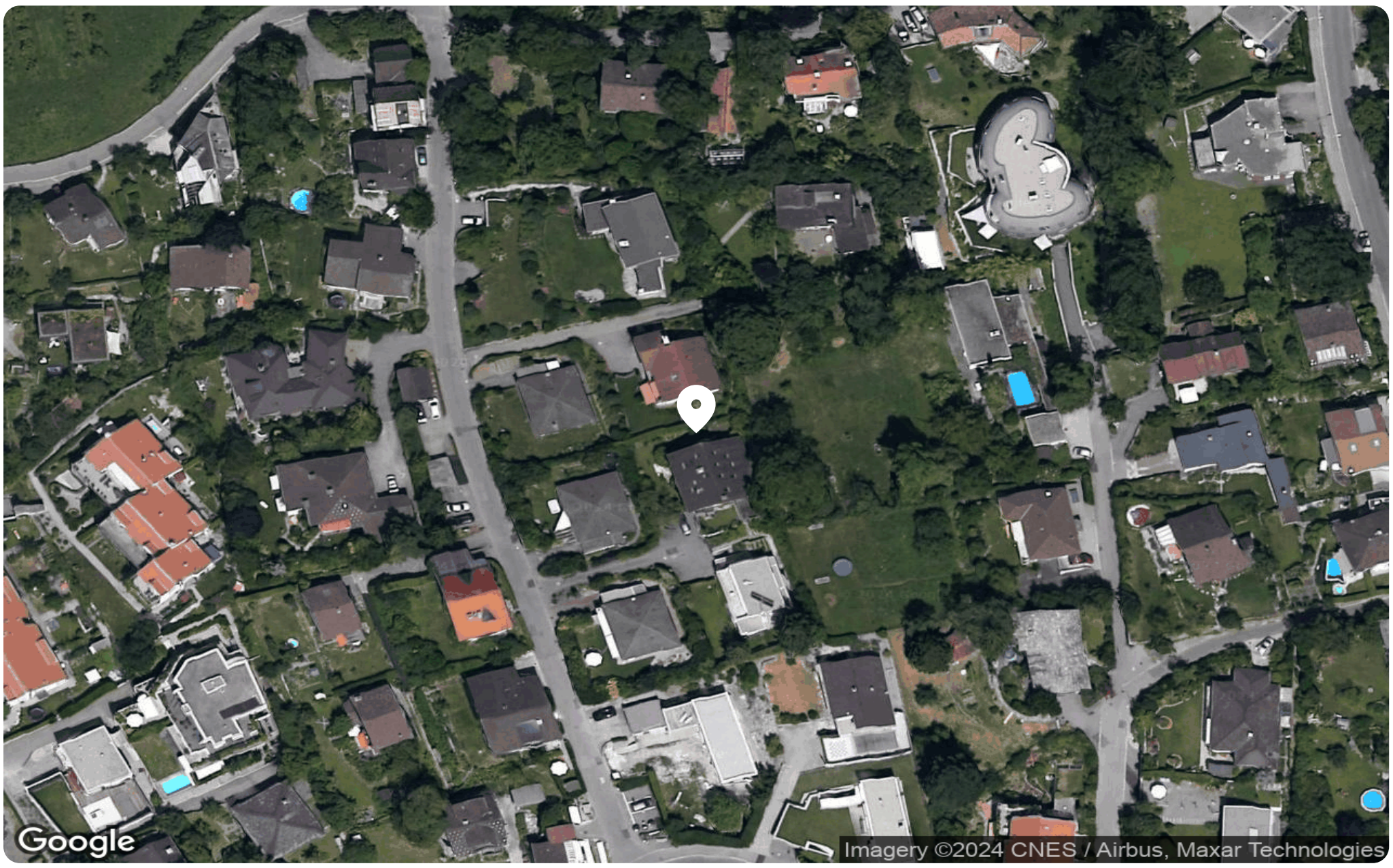
NORM

Energy certificate from 12. February 2024

Plantaweg 6
7000 Chur



Single-family house	305 m²	1967	D Energy class
Building type	Heated Area	Year of construction	

Norm Verified Plan

Intro

Hello Peter Müller.

Welcome to your personal energy report. I hope you have an exciting and informative experience exploring it.



Sandro Fuchs, Your energy expert

Client

Peter Müller

Owner
peter.muellermd@gmail.com
+41 79 408 92 06

Energy expert

Sandro Fuchs
Norm Technologies AG

info@norm.ch
www.norm.ch

Digital energy certificate

Certificate ID:
Xry1gQjBndlgBWIcYd5M

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Assessment

The energy class indicates how much energy your house requires and the amount of CO_2 emissions it produces. This energy demand is displayed in classes from A to G. A distinction is made between the following 3 categories.

Total energy efficiency

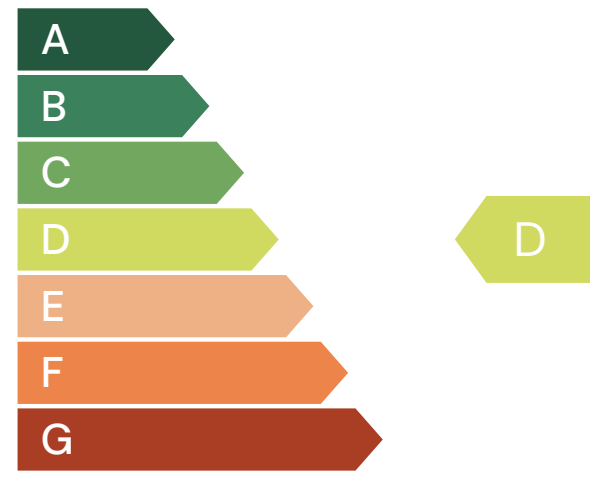
Includes heating demand, building technology (heat generation including hot water), electricity demand, and own electricity production.

Assessment

The energy efficiency (heating, hot water, electricity) is inadequate. It does not meet the current energy requirements by a factor of two.

Energy class

176.9 kWh/m²a



Building envelope efficiency

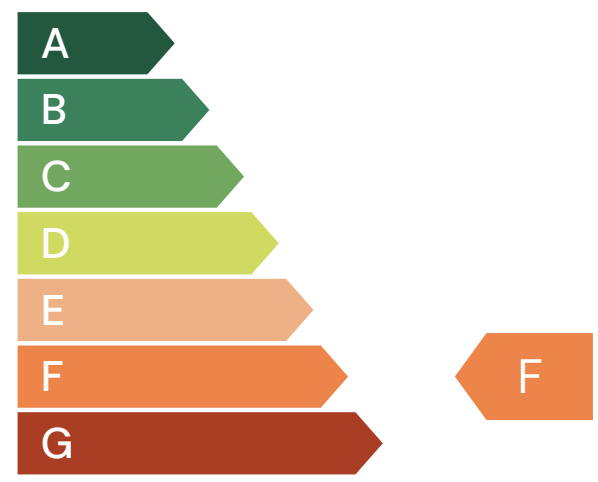
Describes the thermal insulation quality of the building envelope including walls, roof, floors, and windows.

Assessment

The building envelope is poorly insulated. It loses almost three times the energy compared to current new buildings.

Energy class

116.3 kWh/m²a



CO_2 emissions

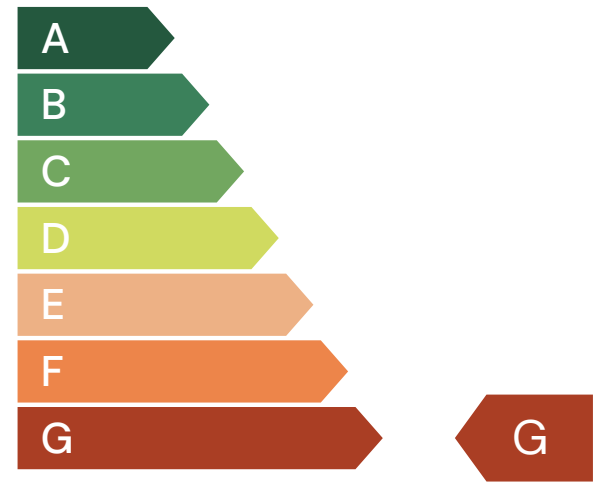
Indicates how much CO_2 is emitted by the building for space heating and hot water.

Assessment

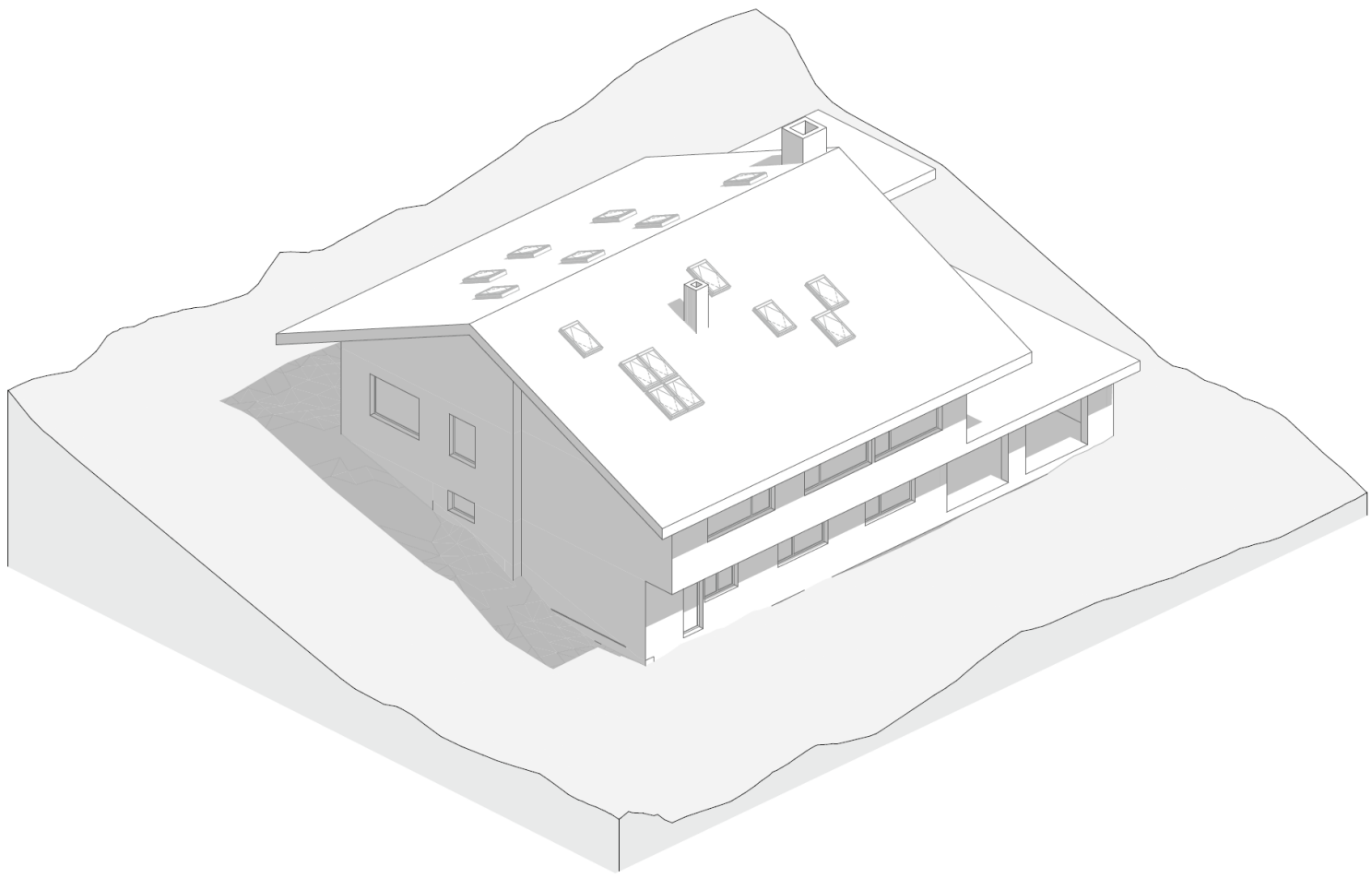
The operation of the building has extremely high CO_2 emissions due to fossil heating.

CO_2 emissions

27.5 kg CO_2/m²a



Facts & Figures



Building	
Building type	Single-family house
Year of construction	1967
Residents	2
Heating	Gas Heating

Surfaces	
Heated Area	305 m²
Number of windows	33
Window area	43 m²
Facade area	191 m²
Roof area	134 m²

Energy consumption	
Energy consumption per year	31'477 KWh
Energy costs per year	4'185 CHF
CO_2 emissions per year	8'388 kg CO_2

State

Describes the current state of all central components.

Roof	Good
Facade	Average
Windows & Doors	Average
Floor	Insufficient
Heating	Average
Hot water	Very good
Devices	Good

Roof

The pitched roof is a flumser roof, constructed similarly with ventilation and visible rafters. The insulation layer of 20cm (possibly in 2 layers) is above the rafter construction, and the roofing is made of clay tiles. The roof extension (previously a flat roof) was built in 1982. The technical lifespan (insulation) of 30 years has been exceeded.

State	Good
Energy loss	7%
Last renovation	1982
Next recommended renovation	2024

Recommended renovation

Roof renovation

Roof renovation with 20cm mineral wool & 3.5cm wood fiber insulation (under-roof), 8cm ventilation gap due to PV system, replacement of roof windows, and removal of roof windows on the west side.

Outer wall; U-value < 0.20 W/m2*K or at least 0.07 W/m2*K better than the existing, must be 0.17 W/m2*K (Building Program)

Costs	45'500 CHF
Subsidies	9'730 CHF

Additional benefits

Soundproofing
Free from noise and sounds from the surroundings

Heat protection
Protection against overheating in summer

Property value
Increase in the market value of the building

Thermal comfort
Free from unpleasant drafts, heat or cold radiation

Healthy living
Free from moisture, mold, and toxins in indoor spaces

Architectural possibilities
Allows the external appearance to be redesigned

Facade

The plastered compact facade was externally insulated with 8 cm in 1982. The remaining walls against the earth, which are unheated, are not thermally insulated. The technical lifespan of 40 years has been exceeded.

State	Average
Energy loss	31%
Last renovation	1982
Next recommended renovation	2024

Recommended renovations

Wall insulation

Insulate walls against unheated spaces with, for example, 12 cm of mineral wool.

Walls facing unheated areas; U-value < 0.28 W/m²·K (MuKEn14 renovation requirement).

Costs	4'500 CHF
Subsidies	-

Facade insulation

External insulation, for example, 16cm mineral wool.

Exterior wall; U-value < 0.20 W/m²K (Building Program)

Costs	45'300 CHF
Subsidies	12'078 CHF

Additional benefits

Property value
Increase in the market value of the building

Thermal comfort
Free from unpleasant drafts, heat or cold radiation

Soundproofing
Free from noise and sounds from the surroundings

Heat protection
Protection against overheating in summer

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Healthy living
Free from moisture, mold, and toxins in indoor spaces

Windows & Doors

The wooden windows are double glazed and were installed in 1982. The balcony door and some roof windows were replaced with triple glazed windows in 2000. The entrance door (wooden) appears to be leaking. The technical lifespan of 25 years has mostly been exceeded.

State	Average
Energy loss	22%
Last renovation	1982
Next recommended renovation	2024

Recommended renovations

Interior doors replacement

Replace doors towards unheated areas.

Interior doors; U-value < 1.5 W/m²K (MuKEn14 renovation requirement)

Costs	22'000 CHF
Subsidies	-

Window and door replacement

Window and door replacement for exterior.

Windows; triple glazing, U-value < 1.0 W/m²·K (MuKEn14 renovation requirement)

Exterior doors; U-value < 1.2 W/m²·K (MuKEn14 renovation requirement)

Costs	37'100 CHF
Subsidies	-

Additional benefits

Soundproofing
Free from noise and sounds from the surroundings

Property value
Increase in the market value of the building

Thermal comfort
Free from unpleasant drafts, heat or cold radiation

Heat protection
Protection against overheating in summer

Security
Protection against burglary and theft

Architectural possibilities
Allows the external appearance to be redesigned

Floor

The floors are made of reinforced concrete, and based on the building's construction year, we assume that a minimal insulation of 2cm was installed in the subfloor. The technical lifespan of 30 years has long been exceeded.

State	Insufficient
Energy loss	29%
Last renovation	1967
Next recommended renovation	2024

Recommended renovations

Basement ceiling insulation

Insulate the underside of the basement ceiling, for instance, with 10 cm of mineral wool.

Ceiling insulation; U-value < 0.28 W/m²·K (MuKEn14 renovation requirement).

Costs	6'300 CHF
Subsidies	-

Floor insulation

New base floor with 2cm TSD & 10cm PIR

Floor against soil ≤ 2 m; U-value < 0.20 W/m²K (Building Program)

Costs	11'500 CHF
Subsidies	3'060 CHF

Additional benefits

Property value
Increase in the market value of the building

Thermal comfort
Free from unpleasant drafts, heat or cold radiation

Architectural possibilities
Allows the external appearance to be redesigned

Heating

The building is heated by a gas heating system installed in 2013. Heat is distributed in the attic through underfloor heating, in the upper floor through radiators, and on the ground floor through a combination of underfloor heating and radiators. The radiators are equipped with old thermostat valves. All visible heating pipes are insulated. The gas heating system has not yet reached its technical lifespan of 20 years.

State	Average
Last renovation	2013
Next recommended renovation	2033

Recommended renovation

Replacement heating and solar thermal system

Total renovation of the heating system including hot water. (Fossil-free)

New thermal solar system (Area approx. 6m2 for 4 persons, approx. 1.5m2 per person)

Costs	76'700 CHF
Subsidies	6'840 CHF

Additional benefits

Property value
Increase in the market value of the building

Hot water

The hot water is heated by a 6m2 (approximately 3.0m2 per person) thermal solar system, which has been in operation since 2013. Generally, a thermal solar system requires 1.5 m2 per person. During peak times, the hot water production is supported by the gas heating system. The heating pipes and hot water lines in the basement are located outside the thermal building envelope and are insulated. The technical lifespan of the hot water tank, which is 20 years, has not yet been reached. The construction year of the thermal solar system is unknown.

State	Very good
Last renovation	2013
Next recommended renovation	2033

Recommended renovation

Replacement heating and solar thermal system

Total renovation of the heating system including hot water. (Fossil-free)

New thermal solar system (Area approx. 6m2 for 4 persons, approx. 1.5m2 per person)

Costs	76'700 CHF
Subsidies	6'840 CHF

Additional benefits

Property value
Increase in the market value of the building

Devices

The lighting fixtures are quite basic. As the building is occupied throughout the day, some of the light bulbs have been replaced with LED energy-saving bulbs. The household appliances are in poor to fair condition. The washing machine and dishwasher are not connected to the hot water supply.

State	Good
Next recommended renovation	2024

Recommended renovations

Electrical appliances replacement

Replace all household appliances, if possible connect washing machine and dishwasher to hot water.

Costs	16'400 CHF
Subsidies	-

PV system installation

PV system on the west-facing roof (76m2). At least 70% of the suitable roof area, combined with roof renovation, qualifies for a bonus of 30.-/m2.

The bonus is already included in the roof renovation.

Costs	40'000 CHF
Subsidies	5'680 CHF

Additional benefits

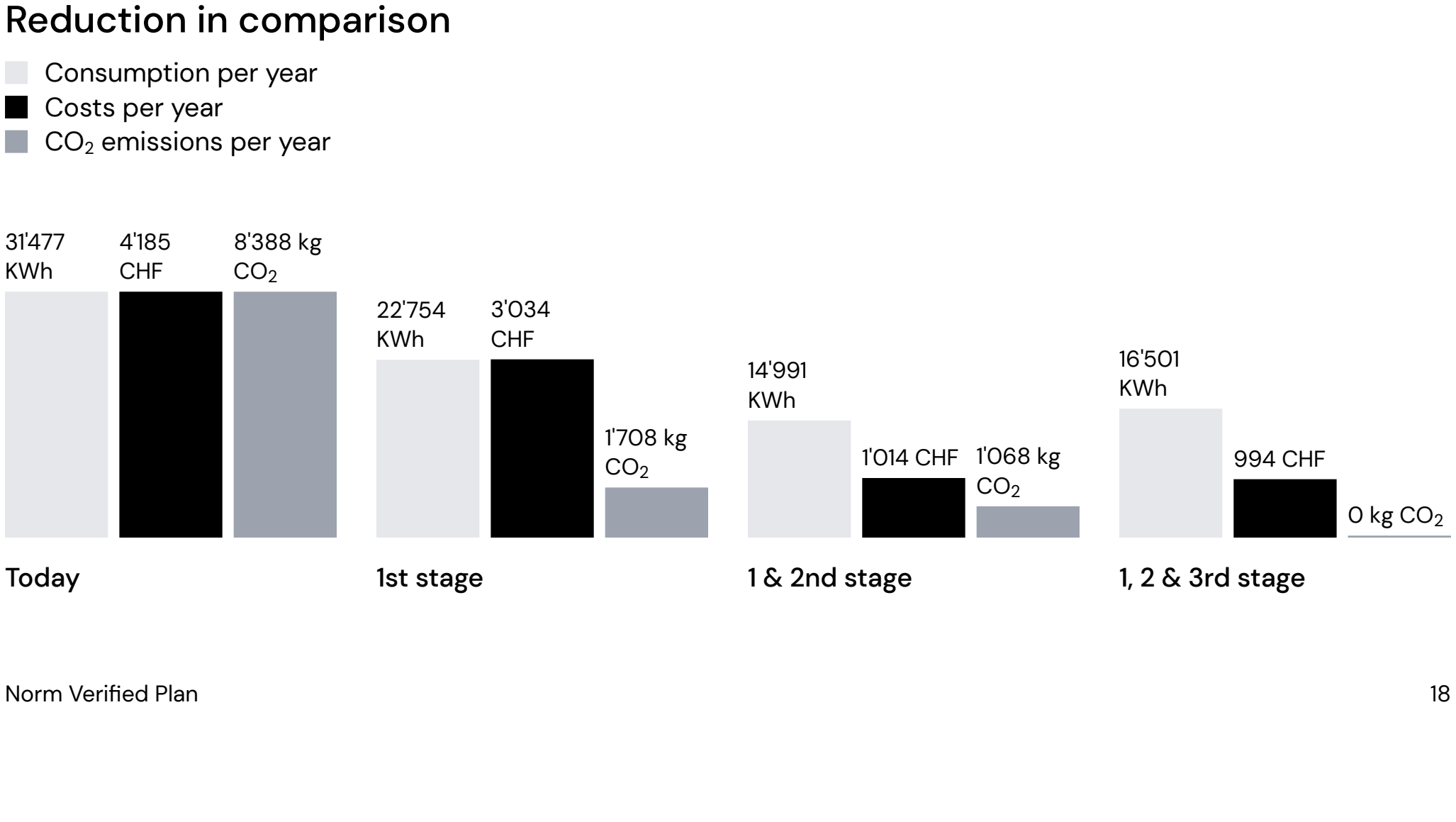
Property value
Increase in the market value of the building

Refurbishment schedule

Planning of refurbishment measures based on the service life and condition of the components.

		① Wall insulation & Basement ceiling in... Short-term measures <1 year	② Window and door replacement & Facade ... Mid-term measures - 1 to 5 years	③ Replacement heating and solar thermal... Long-term measures - 5 to 10 years
Energy				
New consumption per year	22'754 KWh ▼ 28%	14'991 KWh ▼ 52%	16'501 KWh ▼ 48%	
New costs per year	3'034 CHF ▼ 1'151 CHF	1'014 CHF ▼ 3'171 CHF	994 CHF ▼ 3'191 CHF	
CO_2 emissions per year	1'708 kg CO_2	1'068 kg CO_2	0 kg CO_2	
Energy class				
Total energy	C ▲ 1	B ▲ 2	A ▲ 3	
Building envelope	D ▲ 2	B ▲ 4	B ▲ 4	
CO_2 emissions	E ▲ 2	D ▲ 3	A ▲ 6	
Renovation				
Costs	57'640 CHF	140'412 CHF	69'860 CHF	

Reduction in comparison



1

Wall insulation & Basement ceiling insulation & Interior doors replacement & Electrical appliances replacement & Floor insulation

In the short term, it is necessary to renovate the building envelope adjacent to unheated spaces and the ground. Insulation will be added to walls and floors that are next to these unheated areas and the ground to mitigate significant heat loss compared to other parts of the building. Efforts will also be made to minimize thermal bridges at the junctures between heated and unheated spaces. Furthermore, as part of the planned kitchen renovation, all electrical appliances will be replaced simultaneously.

Effects of the 1st stage

Energy		
Consumption per year	▼ 28%	22'754 KWh
Costs per year	▼ 1'151 CHF	3'034 CHF

Energy class		
Total energy	▲ 1	C
Building envelope	▲ 2	D
CO_2 emissions	▲ 2	E

Recommended measures

Basement ceiling insulation

Insulate the underside of the basement ceiling, for instance, with 10 cm of mineral wool.

Ceiling insulation; U-value < 0.28 W/m²·K (MuKEn14 renovation requirement).

Costs	6'300 CHF
Subsidies	-

Electrical appliances replacement

Replace all household appliances, if possible connect washing machine and dishwasher to hot water.

Costs	16'400 CHF
Subsidies	-

Floor insulation

New base floor with 2cm TSD & 10cm PIR

Floor against soil ≤ 2 m; U-value < 0.20 W/m²K (Building Program)

Costs	11'500 CHF
Subsidies	3'060 CHF

Interior doors replacement

Replace doors towards unheated areas.

Interior doors; U-value < 1.5 W/m²K (MuKEn14 renovation requirement)

Costs	22'000 CHF
Subsidies	-

Wall insulation

Insulate walls against unheated spaces with, for example, 12 cm of mineral wool.

Walls facing unheated areas; U-value < 0.28 W/m²·K (MuKEn14 renovation requirement).

Costs	4'500 CHF
Subsidies	-

Costs

Costs	60'700 CHF
Subsidies	3'060 CHF
Tax savings	-
Total	**57'640 CHF**

The figures are estimates. Please obtain quotes to determine the exact costs.

Additional benefits

Property value
Increase in the market value of the building

Thermal comfort
Free from unpleasant drafts, heat or cold radiation

Architectural possibilities
Allows the external appearance to be redesigned

Soundproofing
Free from noise and sounds from the surroundings

2

Window and door replacement & Facade insulation & Roof renovation & PV system installation

In the mid-term, in addition to stage 1 measures, the steep roof and facade, along with window and door replacements, as well as the floor facing outside, will be enhanced with thermal insulation. The west-facing roof windows will be removed to facilitate the installation of a PV system (approximately 12 kWp), which will be carried out in tandem with the roof renovation. These measures are expected to elevate the building to energy class B in terms of both the building envelope and overall energy efficiency.

Effects of the 2nd stage

Energy		
Consumption per year	▼ 52%	14'991 KWh
Costs per year	▼ 3'171 CHF	1'014 CHF

Energy class		
Total energy	▲ 2	B
Building envelope	▲ 4	B
CO_2 emissions	▲ 3	D

Recommended measures

Facade insulation

External insulation, for example, 16cm mineral wool.

Exterior wall; U-value < 0.20 W/m²K (Building Program)

Costs	45'300 CHF
Subsidies	12'078 CHF

PV system installation

PV system on the west-facing roof (76m2). At least 70% of the suitable roof area, combined with roof renovation, qualifies for a bonus of 30.-/m2.

The bonus is already included in the roof renovation.

Costs	40'000 CHF
Subsidies	5'680 CHF

Roof renovation

Roof renovation with 20cm mineral wool & 3.5cm wood fiber insulation (under-roof), 8cm ventilation gap due to PV system, replacement of roof windows, and removal of roof windows on the west side.

Outer wall; U-value < 0.20 W/m2*K or at least 0.07 W/m2*K better than the existing, must be 0.17 W/m2*K (Building Program)

Costs	45'500 CHF
Subsidies	9'730 CHF

Window and door replacement

Window and door replacement for exterior.

Windows; triple glazing, U-value < 1.0 W/m²·K (MuKEn14 renovation requirement)

Exterior doors; U-value < 1.2 W/m²·K (MuKEn14 renovation requirement)

Costs	37'100 CHF
Subsidies	-

Costs

Costs	167'900 CHF
Subsidies	27'488 CHF
Tax savings	-
Total	**140'412 CHF**
Total 1 & 2nd stage	198'052 CHF

The figures are estimates. Please obtain quotes to determine the exact costs.

Additional benefits

Soundproofing
Free from noise and sounds from the surroundings

Heat protection
Protection against overheating in summer

Property value
Increase in the market value of the building

Thermal comfort
Free from unpleasant drafts, heat or cold radiation

Architectural possibilities
Allows the external appearance to be redesigned

Healthy living
Free from moisture, mold, and toxins in indoor spaces

Security
Protection against burglary and theft

3

Replacement heating and solar thermal system

Building on the measures from stages 1 and 2, the existing fossil gas heating system will be dismantled and replaced with a new pellet heating system. Additionally, the current thermal solar system will undergo renewal. It is advised to synchronize the hot water preparation and the operation of the pellet heating system with both the thermal solar system and the PV system, to optimize the self-consumption of electricity.

Effects of the 3rd stage

Energy		
Consumption per year	▼ 48%	16'501 KWh
Costs per year	▼ 3'191 CHF	994 CHF

Energy class		
Total energy	▲ 3	A
Building envelope	▲ 4	B
CO_2 emissions	▲ 6	A

Recommended measures

Replacement heating and solar thermal system

Total renovation of the heating system including hot water. (Fossil-free)

New thermal solar system (Area approx. 6m2 for 4 persons, approx. 1.5m2 per person)

Costs	76'700 CHF
Subsidies	6'840 CHF

Costs

Costs	76'700 CHF
Subsidies	6'840 CHF
Tax savings	-
Total	**69'860 CHF**
Total 1, 2 & 3rd stage	267'912 CHF

The figures are estimates. Please obtain quotes to determine the exact costs.

Additional benefits

Property value
Increase in the market value of the building

Next steps



Personal advice
Ask our experts all your questions. CHF 160 per hour.
info@norm.ch

Apply for subsidies
Order a GEAK Plus to apply for subsidies.
info@norm.ch

Find a craftsman
Find suitable and certified tradespeople via the Houzy platform.
Start request

Check financing
Expert advice on financing your renovation project.
Make an appointment

What is a Verified Plan?

The "Norm Verified Plan" assesses the energy efficiency of your property based on standardized usage scenarios. It provides you with a clear overview of the energy performance and improvement potential of your building in line with the latest technical standards. A renovation plan drawn up especially for you sets out how your building can be modernized in stages, including the associated investment costs and the impact of these measures on energy and operating costs. It will also show you which subsidies are available.

About Norm

Norm Technologies AG is a Swiss company based in Zurich, specialized in the creation of accessible energy certificates and proposals for the energy-efficient refurbishment of buildings.

info@norm.ch
https://www.norm.ch

Energy labels

	Building envelope efficiency	Total energy efficiency	CO_2 emissions
A	The building envelope is excellently insulated. It significantly exceeds current standards for new buildings.	The energy efficiency (heating, hot water, electric) is excellent. It can be compared to the best building standard/Minergie standard.	The operation of the building emits no direct CO_2 emissions.
B	The building envelope is very well insulated. It exceeds or meets current standards for new buildings.	The energy efficiency (heating, hot water, electric) is very good. It meets current energy requirements and is comparable to new buildings.	The operation of the building emits CO_2 emissions only in exceptional situations (e.g., peak performance).
C	The building envelope is acceptably insulated. However, it does not meet current standards for new buildings.	The energy efficiency (heating, hot water, electric) is sufficient. It barely meets current energy requirements.	The operation of the building emits low CO_2 emissions.
D	The building envelope is averagely insulated. It loses almost twice as much energy as current new buildings.	The energy efficiency (heating, hot water, electricity) is inadequate. It does not meet the current energy requirements by a factor of two.	The operation of the building emits considerable CO_2 emissions.
E	The building envelope is below average in insulation. It loses more than twice as much energy as current new buildings.	The energy efficiency (heating, hot water, electricity) is low. It does not meet the current energy requirements by a factor of two.	The operation of the building emits high CO_2 emissions.
F	The building envelope is poorly insulated. It loses almost three times the energy compared to current new buildings.	The energy efficiency (heating, hot water, electric) is very low. It does not meet current energy requirements by more than a factor of two.	The operation of the building has very high CO_2 emissions due to fossil heating.
G	The building envelope is very poorly or not insulated at all. It loses many times the energy compared to current new buildings.	The energy efficiency (heating, hot water, electric) is extremely low. It does not meet current energy requirements by multiple times.	The operation of the building has extremely high CO_2 emissions due to fossil heating.